

**Privacy Interfaces as Corporate Communication: A Computational Auditing Framework for Tracking the
Interface Designs**

Key Words:

Computational Auditing of Privacy Interfaces, Compliance Risk Assessment, Strategic Privacy Communication

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1. Project Background

As firms increasingly rely on user data to make market decisions, data privacy has become a growing concern (Wu, 2025). In response to pressure from regulatory entities, such as GDPR and CCPA, companies have adopted privacy interfaces, usually manifested as cookie banners, to serve as compliance mechanisms for gaining user consent (Matte et al., 2020). Beyond compliance, these interfaces function as **firm communication tools** that disclose firms' data practices and facilitate customer control over such practices. Nevertheless, firms **vary** in how they design privacy interfaces in terms of affording unbiased choice from users' perspectives. Compliance with regulatory guidelines does not necessarily translate into unbiased choice. Thus, a systematic third-party audit that evaluates consent interfaces in terms of unbiased choices is needed. Besides the public policy interests, the proposed project also theorizes the privacy interface as a communication tool for strategically building trust with users, extending the auditing system into a self-governance tool that helps firms align their privacy strategies with their privacy interface designs.

Prior research has conceptualized privacy interfaces through the **Notice-and-Choice** (N+C) framework, focusing on Notice, which discloses firms' data practices, and Choice, which enables user control over these practices, and examining how this framework impacts users' privacy-related decision-making (Warner, 2020). Building on the N+C framework, this project aims to develop an auditing system that quantifies features related to these two dimensions, thereby developing a quantifiable auditing system for evaluating regulatory compliance and unbiased choice. In addition, privacy interfaces are not static digital objects. Firms frequently update their designs, including banner text, option structures, and visual layouts, in response to evolving regulatory guidance and strategic considerations. Therefore, the existing literature, which focuses on static evaluations, provides limited insight into the evolution of firms' privacy communication strategies.

This project addresses these gaps by developing an automated, **longitudinal** auditing system to evaluate interface changes. It generates versioned records, traceable scores, and structured justifications, enabling systematic analysis of how firms' privacy communication strategies evolve over time. Beyond analysis, the system can be further developed to support internal Privacy Impact Assessments and scalable third-party monitoring, promoting transparent and user-centered privacy communication by increasing public awareness of privacy protection.

2. Significance and Contribution of the Research to the Broader Field

This research makes a methodological contribution by designing a two-layer computational architecture for privacy interface evaluation. Building on this foundation, it strengthens privacy governance by enabling continuous oversight. Through version-controlled archiving and multimodal interface fingerprinting, audit scores remain traceable and reproducible. This structure allows regulators to identify compliance risks over time, shifting oversight toward the systematic assessment of communication quality. By integrating this proactive risk management system into Privacy Impact Assessments, firms can detect and correct noncompliant interface features before they escalate into regulatory exposure or erode consumer trust. Compliance thus becomes an ongoing calibration process. Finally, by simplifying complex interface evaluations into interpretable risk indicators, the system contributes to broader awareness of data sovereignty. Over time, cumulative benchmarking across firms encourages sustained improvements in privacy communication standards while supporting more accountable digital governance practices.

3. Research Questions

RQ1: How to develop a computational audit and scoring system to quantify the layered consent interfaces in terms of unbiased choice across the full consent pathway?

In practice, firms deploy layered privacy interfaces in which initial banners provide high-level choices, and more granular preferences are available in secondary layers. By addressing this question, this research aims to quantify the features related to primary and secondary privacy interfaces, assessing them systematically in terms of how current privacy interface designs facilitate or obstruct unbiased choices from users' perspectives.

RQ2: How can we automatically capture and version firms' privacy interfaces to systematically document interface changes over time?

The aim is to develop a method for tracking changes in consent interfaces, resulting in a versioned archive and a change log that summarizes updates (Belcheva et al., 2023). While prior scholars have developed databases of privacy policies (Princeton-Leuven Longitudinal Corpus of Privacy Policies), this research extends that approach by constructing a **longitudinal** database of privacy interfaces, contributing to ongoing dialogue on privacy interface design between academic researchers and business practitioners.

4. Methodology and Research Approach

4.1 RQ1: A Three-Layer Audit Framework for Compliance Risk Assessment

To support proactive compliance risk assessment, this research develops a three-layer computational audit framework to identify design-level compliance risk indicators in layered consent interfaces. The framework emphasizes explainability, traceability, and early risk signaling rather than legal determination.

1) **Layer 1: Path Availability**

Layer 1 evaluates whether all four consent decision pathways (Accept, Reject, Customize, and Dismiss) are present and reachable. If any pathway is missing or inaccessible, the interface is flagged with a high design-level compliance risk indicator, the missing pathway is documented in the audit report, and no further analysis is conducted. This layer functions as a necessary gate, as subsequent evaluations are not meaningful when fundamental consent options are unavailable. This assessment signals elevated audit attention under prevailing consent standards.

2) **Layer 2: Path Effort (Operational Accessibility & Independent Risk Signal)**

For interfaces that satisfy Path Availability, Layer 2 evaluates Path Effort, measuring the operational effort required for users to access each consent pathway. Path Effort is quantified as a continuous score derived from visual and structural features, including button size, color contrast, layout symmetry, click depth, label clarity, and immediate interaction feedback. Machine learning techniques are used for feature extraction and standardization, while risk classification follows predefined rules. The resulting score is mapped into three categories—Easy, Average, and Poor—and functions both as an independent design-level risk signal and as a weighting factor applied to subsequent evaluations of Transparency and Unbiased Choice.

3) **Layer 3: Transparency and Unbiased Choice (Risk Explanation Layer)**

Layer 3 evaluates **Transparency** and **Unbiased Choice** conditional on the outcomes of the first two layers and functions as a risk explanation mechanism that identifies why an interface may present elevated compliance risk after baseline access conditions are met (D’Assergio et al., 2025; Martin & Murphy, 2016).

Transparency is operationalized as a content-level assessment of communicative quality, examining how privacy information is disclosed and framed across interface layers (Betzing et al., 2020). It is evaluated along two independent dimensions. **Disclosure Topic Coverage** assesses whether key categories of information—types of data collected, purposes of data use, data sharing or third-party access, and consequences of consent decisions—are disclosed at the point of initial decision and remain consistent across deeper layers. Each topic is coded based on its presence and clarity

at the first layer and its consistency in subsequent layers. **Communicative Framing Effects** evaluate how factually equivalent information is communicated across layers. It captures framing mechanisms that influence user interpretation, including emphasis patterns, linguistic complexity, sequencing of information, and the selective prominence of benefits versus risks. While different framings may remain factually accurate, certain presentation strategies can increase cognitive effort, obscure key disclosures, or subtly bias interpretation. Persistent use of framing structures that heighten cognitive burden results in a downward adjustment of the Transparency score. Greater weight is assigned to first-layer disclosures given their dominant role in shaping initial user interpretation, and the overall Transparency score is computed as a weighted combination of Disclosure Topic Coverage and Communicative Framing Effects.

Unbiased Choice assesses whether the visual and structural presentation of consent options systematically favors one pathway over others, focusing exclusively on observable asymmetries in visual prominence, layout structure, and pathway balance. By isolating structural choice symmetry from communicative clarity, the framework ensures that Transparency and Unbiased Choice remain analytically distinct.

4.2 RQ2: Longitudinal Database of Privacy Interfaces

An automated web-scraping pipeline will be developed to capture weekly updates to privacy consent interfaces. Since consent banners cannot be traced through the Wayback Machine, a prospective tracking approach will be used, running from the present until spring 2027. The crawler will visit the same websites each week, capturing consent banners and deeper consent layers when available. Each interface will be assigned a multimodal fingerprint (text, screenshots, DOM structure) and compared to previous versions. If differences are detected, changes will be logged and stored in a PostgreSQL database. The system will produce a versioned archive of consent interfaces, a weekly log of interface updates, and multimodal capture bundles (text, screenshots, structural representations) for both first-layer and second-layer interfaces.

5. Time Commitment and Summer Work Plan

I will dedicate 40 hours per week to this project from May 18 to August 14, 2026 (13 weeks). This will be a full-time commitment, and I will not take any other courses during this period. All research objectives outlined in this proposal will be

Weeks	Objectives	Deliverables
Weeks 1–2	Establish baseline infrastructure for longitudinal auditing	Configured scraping pipeline; finalized site list and consent scope; baseline capture of first- and second-layer consent interfaces; initialized database and storage structure.
Weeks 3–4	Enable systematic version tracking of consent interfaces	Operational weekly monitoring workflow; multi-modal interface fingerprints (text, screenshots, DOM); validated change detection and version logging.
Weeks 5–6	Implement core audit framework components	Deployed Layer 1 (Path Availability) and Layer 2 (Path Effort) metrics; defined scoring logic and thresholds; preliminary audit results on sampled interfaces.
Weeks 7–8	Operationalize transparency and choice evaluation	Implemented Layer 3 Transparency and Unbiased Choice indicators; evaluated disclosure consistency and structural symmetry across consent layers.
Weeks 9–10	Generate analyzable research outputs	Longitudinal analysis of interface changes; comparative audit summaries; development of representative case studies.
Weeks 11–12	Prepare publishable research materials	Drafted full research paper; prepared figures, tables, and audit visualizations; designed poster content.
Week 13	Finalize and package research outputs	Final paper and poster ready for submission; documented system architecture and workflow for post-summer extension.

completed and delivered in the form of a research paper and a poster. After the summer, I will continue to build on this foundation through my graduation in Spring 2027 by refining specific methodological components, adding more granular indicators, and further developing the framework with a stronger focus on regulatory compliance.

6. Expected Learning Outcomes and Career Goals

Through this project, I aim to develop as a data scientist with strong business awareness who treats privacy compliance as a default constraint in real-world analytics work. With the growing emphasis on data lineage, auditability, and regulatory accountability, privacy considerations increasingly shape how data systems are deployed. By building a longitudinal auditing system for consent interfaces, I will strengthen my technical skills in multimodal data pipelines and interface analysis while learning how governance and risk signals can be translated into measurable, traceable indicators. This foundation supports my early career interests in roles that sit between data science and governance. More broadly, the project frames privacy as an operational issue that can be communicated and improved with evidence, contributing to clearer industry practices and greater public understanding.

7. Budget Summary

I am requesting \$4,000 from the SOURCE office to support the summer execution of this research project. The funding will support the technical infrastructure required to build and deploy the first operational version of the longitudinal privacy interface auditing system.

Item	Description	Cost (USD)
Cloud Virtual Server Hosting	Dedicated cloud instance to run weekly automated crawling and multi-layer interface capture	1600
Managed PostgreSQL Database Hosting	Cloud-based database service for version-controlled storage of interface records and audit scores	900
Secure Cloud Storage (Backups)	Encrypted object storage for screenshots, DOM captures, and weekly archive backups	800
Server Monitoring & Logging Services	Runtime monitoring, automated job scheduling, and log retention to ensure reproducibility	500
Infrastructure Configuration & Deployment Support	Initial server setup, environment configuration, deployment testing, and summer maintenance	200
TOTAL		4000

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